

Association Rules

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Association Rules

Association rule mining is a data analysis technique used to identify patterns between items that are frequently purchased together (Pierson, 2019). In this analysis, the Cosmetics dataset (Figure 1) from a large chain drugstore was examined to uncover purchasing patterns that could support improved point-of-sale displays, cross-selling strategies, and future electronic recommendation systems.

Figure 1

Preview of Cosmetic Dataset

Data Format

```
[3]: address = 'C:/Users/schwe/Downloads/Cosmetics.csv'
      basket = pd.read_csv (address)
```

```
[4]: basket.head()
```

	Bag	Blush	Nail Polish	Brushes	Concealer	Eye brow Pencils	Bronzer	Lip liner	Mascara	Eye shadow	Foundation	Lip Gloss	Lipstick	Eyeliner
0	0	1	1	1	1	0	1	1	1	0	0	0	0	1
1	0	0	1	0	1	0	1	1	0	0	1	1	0	0
2	0	1	0	0	1	1	1	1	1	1	1	1	1	0
3	0	0	1	1	1	0	1	0	0	0	1	0	0	1
4	0	1	0	0	1	0	1	1	1	1	0	1	1	0

Note. Each row represents a single customer transaction, and each column represents a specific product. A value of 1 indicates that the product was purchased in that transaction, while a value of 0 indicates that product was not purchased.

Combinations of Typically Purchased Items

The Apriori algorithm was applied using a minimum support threshold of 2% in order to focus on product combinations that occur with meaningful frequency. A lower threshold was explored (min_support=0.002), but it generated a large number of rare itemsets that were less

practical for analysis. Additionally, by filtering for itemsets with a length ≥ 2 , the analysis concentrated specifically on product relationships.

The analysis identified several product combinations that frequently occur together within customer transactions. As shown in Figure 2, the strongest 2-item combination is Foundation and Lip Gloss, with a support value of 0.356. This indicates that approximately 35.6% of all transactions include both products. Similarly, Eye Shadow and Mascara appear together in 32.1% of transactions, and Eyeliner and Concealer occur together in 29.7% of purchases. These relatively high support values suggest that customers commonly purchase complementary cosmetic products within the same transaction rather than buying individual items in isolation.

Figure 2

Top 2-Item Combinations

Support Calculation

```
[5]: frequent_itemsets = apriori(basket.astype('bool'), min_support=0.02, use_colnames=True)
frequent_itemsets['length'] = frequent_itemsets['itemsets'].apply(lambda x: len(x))
frequent_itemsets
```

[5]:

	support	itemsets	length
0	0.054	(Bag)	1
1	0.363	(Blush)	1
2	0.280	(Nail Polish)	1
3	0.149	(Brushes)	1
4	0.442	(Concealer)	1
...	...	...	...
1458	0.020	(Eye shadow, Concealer, Mascara, Bronzer, Eyel...	7
1459	0.022	(Blush, Eye shadow, Nail Polish, Concealer, Ma...	8
1460	0.021	(Blush, Eye shadow, Nail Polish, Concealer, Ma...	8
1461	0.021	(Foundation, Blush, Eye shadow, Concealer, Mas...	8
1462	0.021	(Eye shadow, Nail Polish, Concealer, Mascara, ...	8

1463 rows × 3 columns

```
[6]: frequent_itemsets[frequent_itemsets['length'] >= 2].sort_values(by='support', ascending=False).head(10)
```

[6]:

	support	itemsets	length
90	0.356	(Foundation, Lip Gloss)	2
81	0.321	(Eye shadow, Mascara)	2
64	0.297	(Eyeliner, Concealer)	2
92	0.238	(Eyeliner, Foundation)	2
61	0.231	(Concealer, Foundation)	2
94	0.227	(Eyeliner, Lip Gloss)	2
27	0.220	(Concealer, Blush)	2
62	0.218	(Concealer, Lip Gloss)	2
86	0.211	(Eye shadow, Foundation)	2
59	0.204	(Concealer, Mascara)	2

Note. The Apriori algorithm was applied with a minimum support threshold of 2%. Only itemsets with a length of ≥ 2 are displayed. The data was then sorted in descending order by support.

Confidence Analysis

Confidence measures the probability that a consequent product is purchased given that the antecedent product or product combination has already been purchased (Pierson, 2019). In

other words, it evaluates how reliably one item predicts the purchase of another, as shown in Figure 3.

The strongest rule identified is {Brushes} $\rightarrow$ {Nail Polish}, with a confidence value of 1.0 and a support value of 0.149. This indicates that 100% of transactions containing brushes also include nail polish, and that this pairing appears in 14.9% of all transactions. This suggests that nail polish is strongly associated with customers who purchase cosmetic application tools.

The second strongest rule further reinforces this pattern: {Brushes, Bronzer} $\rightarrow$ {Nail Polish} also demonstrates a confidence value of 1.0 with support of 0.097. This finding is particularly important because it illustrates how a strong 2-item relationship can expand into a 3-item combination. In this case, brushes serve as the core predictive item. When bronzer is added to the basket, nail polish remains a guaranteed complementary purchase. This progression from a 2-item antecedent to a 3-item antecedent highlights how frequent itemsets can evolve and suggests layered purchasing behavior among customers assembling multiple cosmetic products.

A similar rule, {Concealer, Brushes} $\rightarrow$ {Nail Polish}, also shows a confidence value of 1.0 with support of 0.092. This suggests that the relationship is occurring at a relatively high concept level rather than at specific item variations (Kantardzic, 2019). Customers appear to be purchasing complementary cosmetic categories as part of the same transaction, reinforcing the pattern that nail polish is a consistent add-on when makeup application tools and complexion products are selected.

Figure 3

Top Association Rules Based on Confidence

Confidence

```
[8]: rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.5).sort_values(by=['confidence', 'support'], ascending=[False, False])
rules.head(10)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulc
16	(Brushes)	(Nail Polish)	0.149	0.280	0.149	1.0	3.571429	1.0	0.107280	inf	0.846063	0.532143	1.0	0.0
192	(Bronzer, Brushes)	(Nail Polish)	0.097	0.280	0.097	1.0	3.571429	1.0	0.069840	inf	0.797342	0.346429	1.0	0.0
189	(Concealer, Brushes)	(Nail Polish)	0.092	0.280	0.092	1.0	3.571429	1.0	0.066240	inf	0.792952	0.328571	1.0	0.0
199	(Mascara, Brushes)	(Nail Polish)	0.083	0.280	0.083	1.0	3.571429	1.0	0.059760	inf	0.785169	0.296429	1.0	0.0
203	(Brushes, Eye shadow)	(Nail Polish)	0.081	0.280	0.081	1.0	3.571429	1.0	0.058320	inf	0.783460	0.289286	1.0	0.0
212	(Eyeliner, Brushes)	(Nail Polish)	0.078	0.280	0.078	1.0	3.571429	1.0	0.056160	inf	0.780911	0.278571	1.0	0.0
206	(Brushes, Foundation)	(Nail Polish)	0.074	0.280	0.074	1.0	3.571429	1.0	0.053280	inf	0.777538	0.264286	1.0	0.0
1059	(Mascara, Brushes, Eye shadow)	(Nail Polish)	0.074	0.280	0.074	1.0	3.571429	1.0	0.053280	inf	0.777538	0.264286	1.0	0.0
549	(Nail Polish, Blush, Eye shadow)	(Mascara)	0.071	0.357	0.071	1.0	2.801120	1.0	0.045653	inf	0.692142	0.198880	1.0	0.0
83	(Blush, Brushes)	(Nail Polish)	0.070	0.280	0.070	1.0	3.571429	1.0	0.050400	inf	0.774194	0.250000	1.0	0.0

Note. Association rules are sorted first in descending order by confidence and then by support.

Lift Analysis

Unlike the confidence analysis, the lift results (Figure 4) were not further sorted because the rules were already filtered using minimum thresholds of lift ≥ 5 and confidence ≥ 0.5 . This filtering step ensured that only statistically strong and practically meaningful associations were displayed. Since lift does not have an upper bound like confidence (capped at 1.0), sorting purely by lift prioritized rare combinations with very low support.

One meaningful rule is {Nail Polish, Bronzer} $\rightarrow$ {Brushes}, which has a support of 0.097, confidence of 0.776, and lift of 5.21. This means that nearly 10% of transactions include this combination, and customers who purchase nail polish and bronzer are over five times more likely to also purchase brushes. This suggests that cosmetic products and application tools are frequently purchased together.

Another rule, {Blush, Brushes} -> {Nail Polish, Concealer}, has support of 0.054, confidence of 0.771, and lift of 5.18. This indicates that when customers purchase blush and brushes, they often add nail polish and concealer as well. The lift value shows that this grouping occurs far more often than random purchasing behavior would predict, suggesting consistent cross-category buying patterns.

A more complex rule, {Concealer, Mascara, Lip Liner, Brushes} -> {Nail Polish, Eye Shadow, Bronzer}, has a much higher lift of approximately 15.79 but lower support near 0.021. Although this represents a strong statistical association, it appears in only about 2% of transactions. This highlights an important point: very high lift values may reflect strong but less common purchasing combinations.

Figure 4

Lift Association Rules

Lift and Confidence

```
[16]: rules[(rules['lift'] >= 5) & (rules['confidence'] >= 0.5)]
```

[16]:

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	k
552	(Bronzer, Nail Polish)	(Brushes)	0.125	0.149	0.097	0.776000	5.208054	1.0	0.078375	3.799107	0.923417	0.548023	0.736780	
557	(Brushes)	(Bronzer, Nail Polish)	0.149	0.125	0.097	0.651007	5.208054	1.0	0.078375	2.507212	0.949459	0.548023	0.601151	
1417	(Blush, Brushes)	(Nail Polish, Concealer)	0.070	0.149	0.054	0.771429	5.177373	1.0	0.043570	3.723125	0.867583	0.327273	0.731408	
1424	(Bronzer, Nail Polish, Blush)	(Brushes)	0.069	0.149	0.052	0.753623	5.057874	1.0	0.041719	3.454059	0.861749	0.313253	0.710486	
1433	(Blush, Brushes)	(Bronzer, Nail Polish)	0.070	0.125	0.052	0.742857	5.942857	1.0	0.043250	3.402778	0.894334	0.363636	0.706122	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
39185	(Mascara, Lip liner, Brushes, Eye shadow)	(Eyeliner, Bronzer, Concealer, Nail Polish)	0.034	0.059	0.021	0.617647	10.468594	1.0	0.018994	2.461077	0.936311	0.291667	0.593674	
39186	(Bronzer, Mascara, Lip liner, Brushes)	(Eyeliner, Concealer, Nail Polish, Eye shadow)	0.035	0.053	0.021	0.600000	11.320755	1.0	0.019145	2.367500	0.944732	0.313433	0.577614	
39188	(Bronzer, Lip liner, Brushes, Eye shadow)	(Eyeliner, Mascara, Concealer, Nail Polish)	0.034	0.057	0.021	0.617647	10.835913	1.0	0.019062	2.466308	0.939663	0.300000	0.594536	
39236	(Mascara, Lip liner, Brushes)	(Concealer, Eyeliner, Nail Polish, Eye shadow, ...)	0.038	0.035	0.021	0.552632	15.789474	1.0	0.019670	2.157059	0.973666	0.403846	0.536406	
39239	(Lip liner, Brushes, Eye shadow)	(Concealer, Eyeliner, Nail Polish, Mascara, Br...)	0.038	0.039	0.021	0.552632	14.170040	1.0	0.019518	2.148118	0.966142	0.375000	0.534476	

2393 rows x 14 columns

Note. Rules shown have lift values ≥ 5 and confidence values ≥ 0.5 , highlighting strong and reliable product associations.

Key Takeaways

The results of the association rule analysis provide several actionable insights for the drugstore. First, cosmetic application tools (particularly brushes) consistently appear in strong rules alongside nail polish and other complexion products. This suggests that brushes function as a gateway product, often leading to additional cosmetic purchases. The drugstore could leverage this by placing brushes near nail polish and bronzer displays, training sales associates to recommend complementary products when customers purchase application tools, or even offering targeted promotions on brushes to encourage additional nail product purchases. For example, holding sales or discounts on brushes may indirectly increase nail polish sales by attracting customers into a purchasing pattern that frequently includes both items.

Second, the lift analysis suggests an opportunity to create strategic bundle packages. One of the higher-lift but lower-support rules indicated that customers purchasing concealer, mascara, lip liner, and brushes frequently also purchased nail polish, eye shadow, and bronzer. While this combination occurs in a smaller percentage of transactions, the high lift value suggests a strong purchasing relationship among these broader product categories. This presents an opportunity for the drugstore to test bundled cosmetic sets. For example, offering a bundled package of concealer, mascara, lip liner, and brushes could naturally encourage customers to also purchase a complementary bundle consisting of nail polish, eye shadow, and bronzer. Even if the rule applies to a smaller customer segment, targeted promotions or seasonal kits could capitalize on this structured purchasing behavior.

At the same time, not all product categories demonstrated strong associations. For example, items such as cosmetic bags appeared less frequently in high-confidence or high-lift rules. This suggests that some products may function more as standalone or accessory purchases rather than as core drivers of basket expansion. Recognizing which categories do not strongly influence additional purchases is just as important as identifying strong associations, as it helps the drugstore avoid overemphasizing items that do not meaningfully contribute to cross-selling.

References

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